Year 13 Further Maths - Applications of calculus

Similar questions - Mean value and volumes of revolution

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Find the mean value of $f(x) = 3x^2 - 2x$ over the interval $[0,3]$. **[2]**

2 O is the origin; A, B, C and D have coordinates $(3,0)$, $(3,3)$, $(5,7)$ and $(0,7)$ (units are cm). The arc BC is part of the circle $x^2 + (y-7)^2 = 25$. The shape $OABCD$ is bounded by the line segments OA, AB, the arc BC, and the line segments CD and DO. A funnel is modelled by rotating this shape through 2π about the y-axis. Find the exact volume of the funnel according to the model. **[5]**

3 The region R is bounded by the curve $y = \dfrac{1}{\sqrt{4x^2 - 4x + 2}}$, the x-axis and the lines $x = \frac{1}{2}$ and $x = 1$. The solid S is formed by rotating R through 2π about the x-axis. Show that the volume of S is $\dfrac{\pi^2}{8}$. **[5]**

Question 1

$\frac{1}{3-0}\int_0^3 (3x^2-2x)\,\mathrm{d}x = \frac{1}{3}\left[x^3-x^2\right]_0^3 = \frac{1}{3}(27-9) = 6$

Question 2

The cylinder from $y=0$ to $y=3$ has radius 3: volume $\pi\cdot 3^2\cdot 3 = 27\pi$. For $3 \leqslant y \leqslant 7$, $x^2 = 25-(y-7)^2$. $V_2 = \pi\int_3^7 \left(25-(y-7)^2\right)\mathrm{d}y = \pi\left[25y - \frac{(y-7)^3}{3}\right]_3^7 = \pi\left(175-75-\frac{64}{3}\right) = \frac{236\pi}{3}$. Total $= 27\pi + \frac{236\pi}{3} = \dfrac{317\pi}{3}$ cm^3

Question 3

$V = \pi\int_{1/2}^1 \dfrac{\mathrm{d}x}{4x^2-4x+2}$. $4x^2-4x+2 = 4\left(x-\frac{1}{2}\right)^2+1 = (2x-1)^2+1$. So $V = \pi\left[\frac{1}{2}\tan^{-1}(2x-1)\right]_{1/2}^1 = \pi\cdot\frac{1}{2}\left(\frac{\pi}{4}-0\right) = \dfrac{\pi^2}{8}$